

Introduction to Computer Organisation and Architecture UCCD1133



Chapter 3 Basic Concept of Logic

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Outline

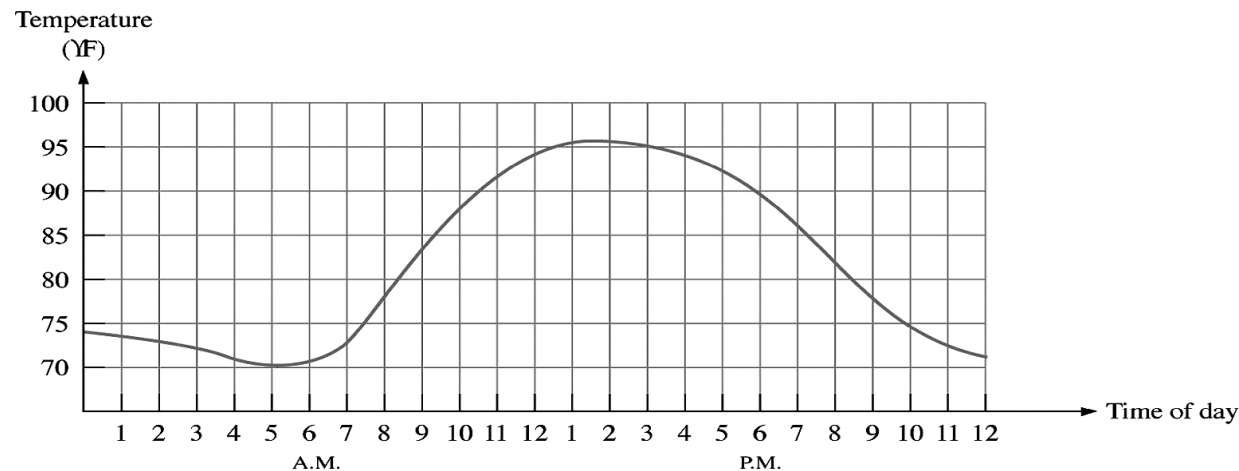
- Analogue and digital quantities
- Binary digits, signal voltage level and logic level
- Data transfer: serial and parallel
- Basic Logic gates

Digital versus Analogue

- Electronic circuits can be divided into two broad categories
 - digital
 - Analogue
- Most physical quantities are analogue in nature.
- However, the electronics inside a computer are digital.

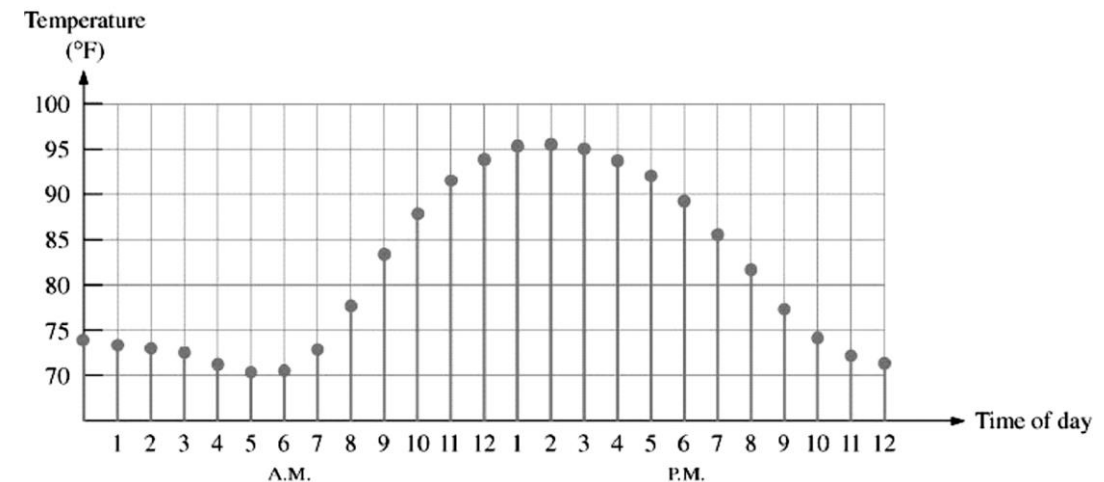
Digital versus Analogue

- An analogue quantity - having continuous values over time
- Example: position, velocity, acceleration, force, pressure, temperature and flow rate.



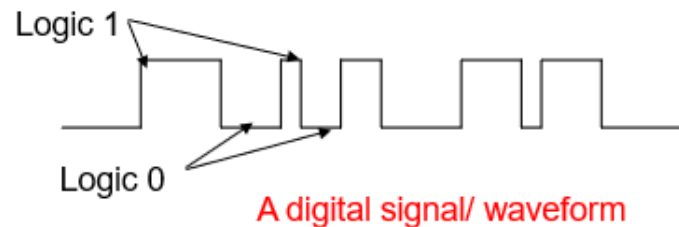
- No sudden jump in value, e.g. from 92 to 94 degrees.
- In between values are considered as infinite.

- An digital quantity - having discrete set of values over time
- Example, the air temperature values are read at every hour – at discrete point in time.
 - Can assign a digital value or code to each dot to represent it - digitised.
- Digital representations of physical quantities are easier to be stored, transferred, and copied.



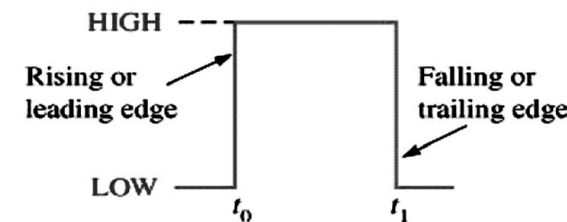
Binary Digits, Signal voltage level and logic level

- Binary system contains only two digits, 0 and 1
 - Bits comes from Binary digits – 0 and 1.
- When apply in digital electronics, 0 and 1 correspond to logic 0 and logic 1

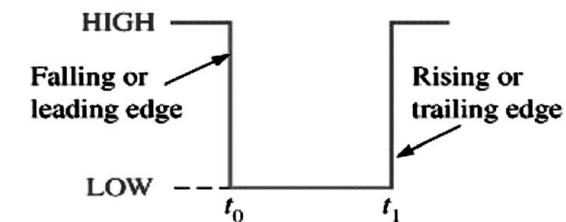


- Voltage levels changing back and forth between the HIGH (H) and LOW (L) instantaneously

- (a) The digital signal is normally LOW, then a positive-going pulse takes place.
- (b) The signal is normally HIGH, then a negative-going pulse takes place.



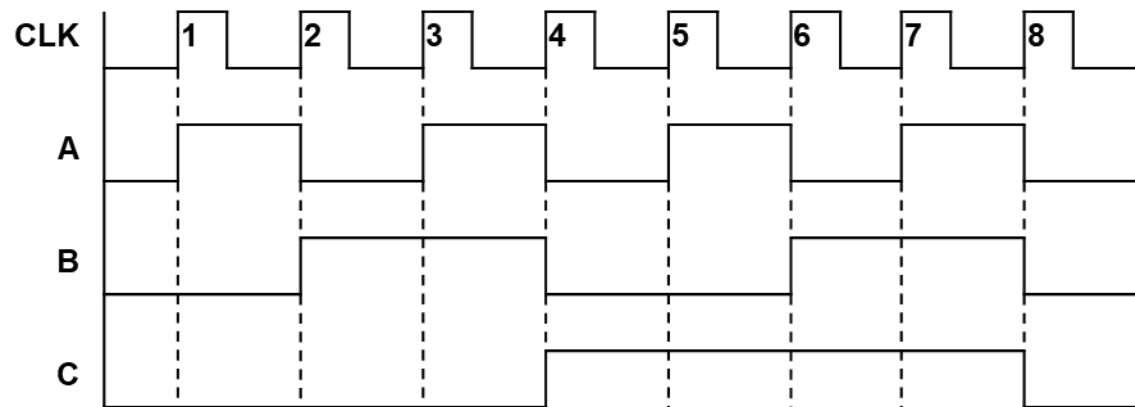
(a) Positive-going pulse



(b) Negative-going pulse

A Special Digital Signal: The Clock Signal

- Clock circuitry is commonly used in digital systems (analogous to a human heart) to generate periodic clock waveform.
 - To synchronize the generation of other waveforms
 - Thus, synchronizing the transfer activities within the digital system.
 - ✓ Creates synchronous digital system
- E.g. waveforms A, B and C are synchronized to the clock.
 - The transitions in waveforms A, B and C occur at the rising-edge of the clock

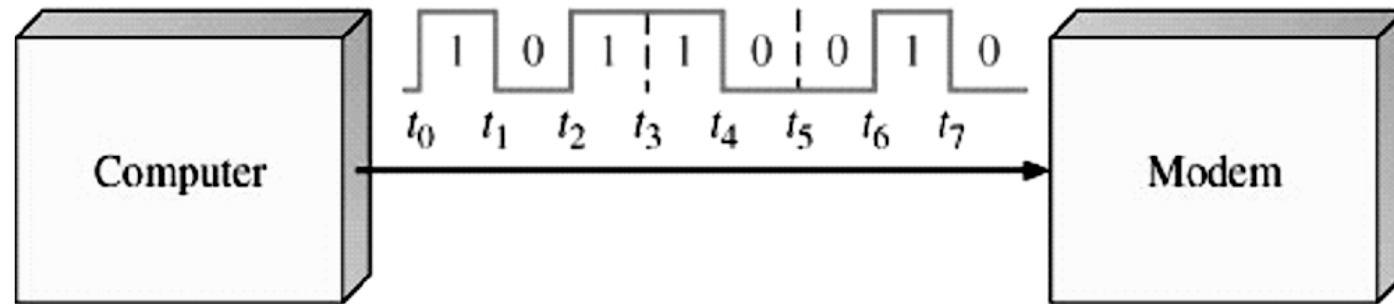


Data Transfer

- Grouping of bits is commonly done to represent information in digital system
 - 4 bits => a nibble
 - 8 bits => a byte
 - 32 bits => a word
 - E.g. 2^8 can represent up to 256 information
- Single or group of bits are transferred as binary data from one circuit to another
 - Serial transfer
 - Parallel transfer
- Data transfer involves a transmitting circuit (source) and one or more receiving circuit (destination)

Serial Data Transfer

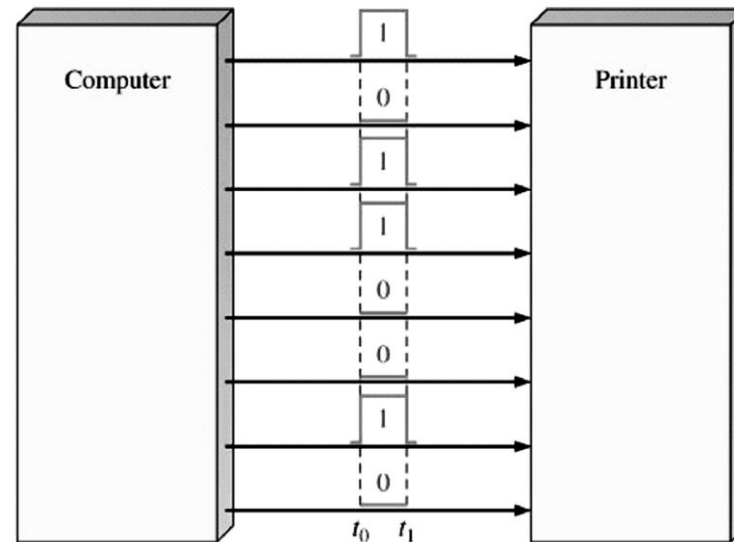
- Data is transferred one bit per clock cycle
 - Require only a single line
- Example, from one digital system (computer) to another (modem)
 - Total clock cycle required to transfer a byte of data: 8 cycles.



(a) Serial transfer of 8 bits of binary data from computer to modem. Interval t_0 to t_1 is first.

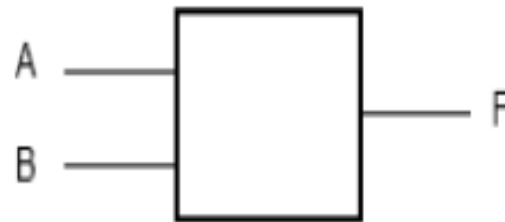
Parallel Data Transfer

- Data is transferred as a group (e.g., a byte or word) per clock cycle
 - Require multiple lines
- Example, from the computer to a printer
 - 8 bits are transferred along 8 lines in a clock cycle
 - Total clock cycle required no matter how large the group: 1 cycle.



Basic Logic Gate

- **Logic gate is fundamental building blocks for all digital circuits.**
- Each gate **at least have one input and only one output.**
- At any given moment, the input and the output is in **one of the two binary** conditions;
 - ✓ **low; 0 ; False** or
 - ✓ **high; 1; True**
- The output depends on the function of the gate and combination of all its inputs.



A generic two –input logic gate

Basic Logic Gate

- **The basic gates are :**
 - i. NOT/INVERTER gate
 - ii. AND gate
 - iii. OR gate
 - iv. NAND gate
 - v. NOR gate
 - vi. XOR gate
 - vii. XNOR gate
 - viii. Negative OR
 - ix. Negative AND

Truth table

- A truth table is a table that describe the behaviour of a logic gates .
- The number of input combinations is equal to 2^N rows for an N-input truth table.

Example 1:

No of input(s) : 2

Number of input combinational(s) entries : $2^N = 2^2 = 4$

Input		Output
A	B	X
0	0	0
0	1	1
1	0	1
1	1	1

Example 2:

No of input(s) : 3

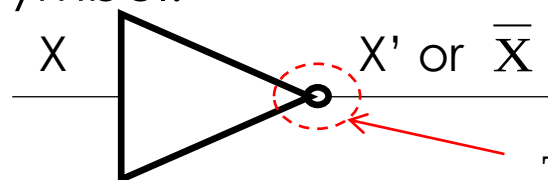
Number of input combinational(s) entries : $2^N = 2^3 = 8$

Input			Output
A	B	C	X
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

Basic Logic Gates (NOT/INVERTER)

- Only inverts the input digital signal (Boolean complementation)

- Symbol:



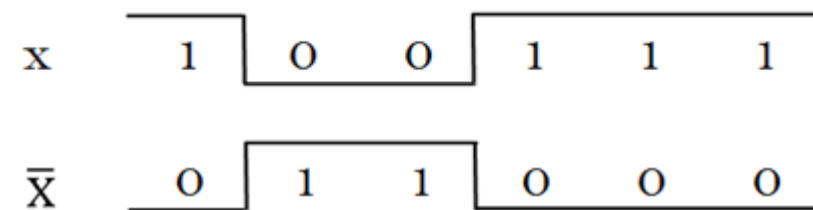
The bubble indicates inversion

- The output is indicated by a prime mark (') or an "elbow" ($\bar{}$).

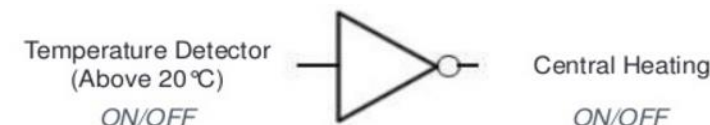
- Truth table:

Input	Output
x	$\bar{x}$
0	1
1	0

- Timing diagram:



- Application of NOT a gate



If the temperature is above 20°C then the Central Heating is switched off.

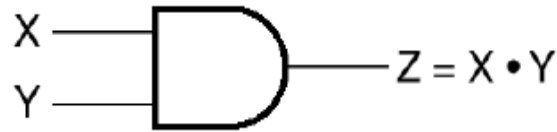
If the temperature is below 20°C then the Central Heating is switched on.

As a detector

Basic Logic Gates (AND gate)

- To represent a product operation, denoted by “.” (Boolean Product)

- Two or more inputs and single output
- Symbol:



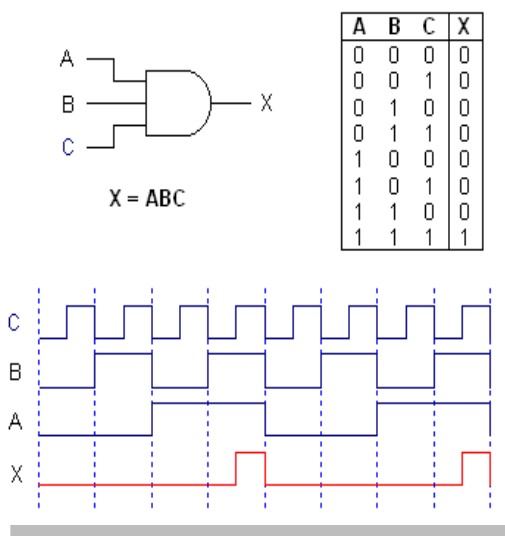
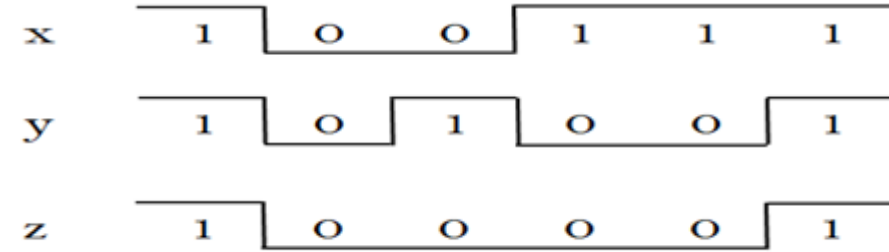
- Truth table:

Input		Output
x	y	z
0	0	0
0	1	0
1	0	0
1	1	1

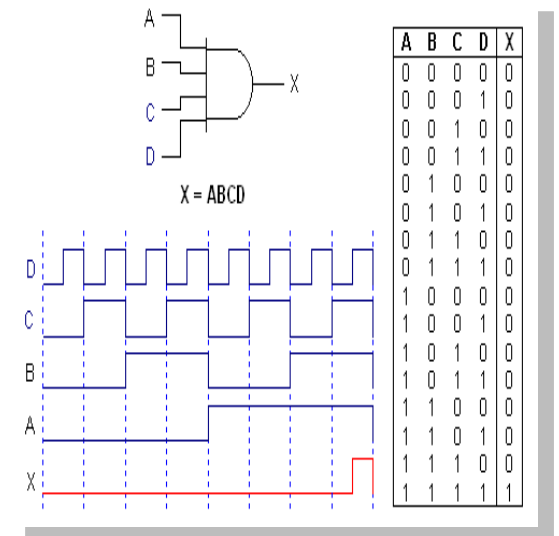
SUMMARY :

- The output is **high (1)** if all the inputs are **high (1)**
- The output is **low (0)** if one or more inputs are **low (0)**

- Timing diagram

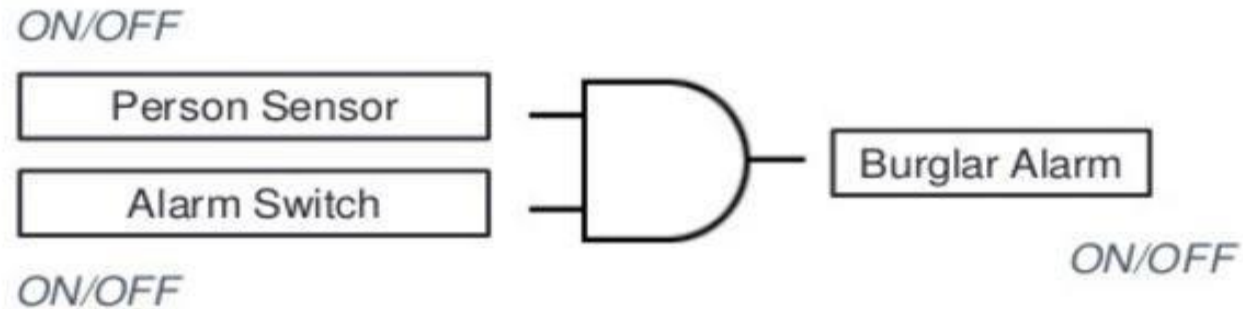


3-Input AND Gate



4-Input AND Gate

Application of an AND gate



If both the Person Sensor AND the Alarm Switch are on then the Burglar Alarm is activated.

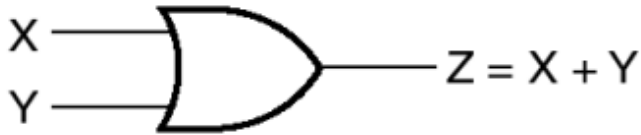


AND gate in a house alarm system

Basic Logic Gates (OR gate)

- To represent a sum operation, denoted by “+” (Boolean Sum)

- Symbol:



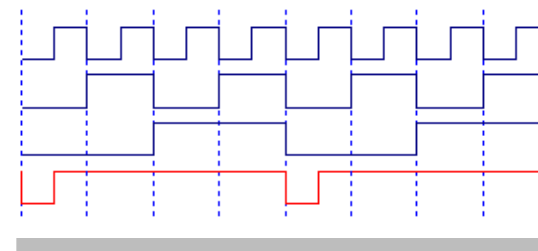
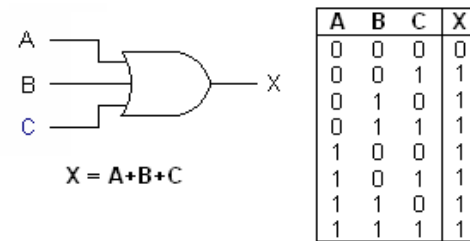
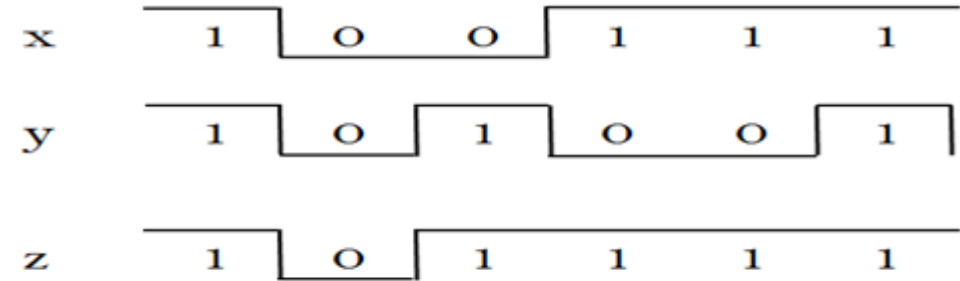
- Truth table:

Input		Output
x	y	z
0	0	0
0	1	1
1	0	1
1	1	1

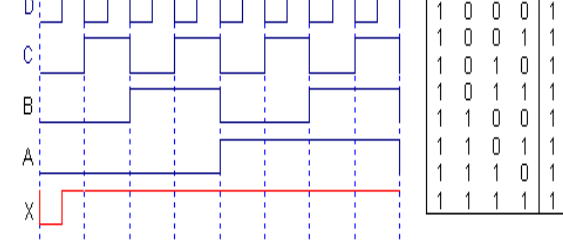
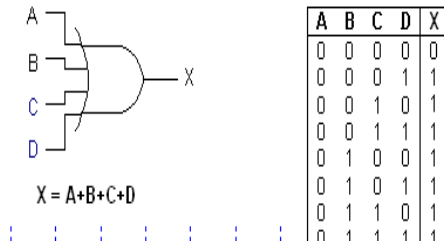
SUMMARY :

- The output is **high (1)** if one or more inputs are high (1)
- The output is **low (0)** if all inputs are low (0)

- Timing diagram

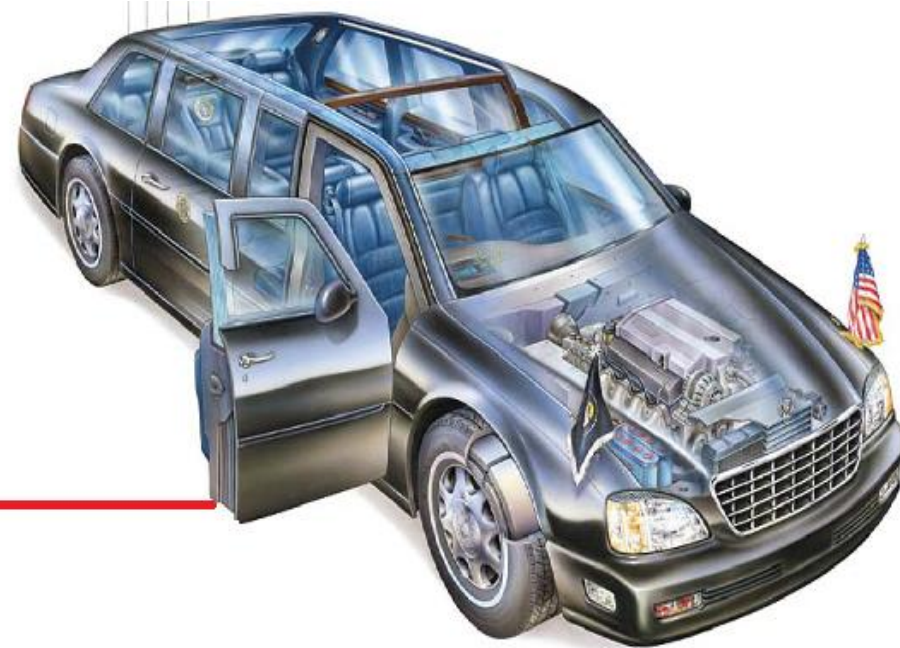
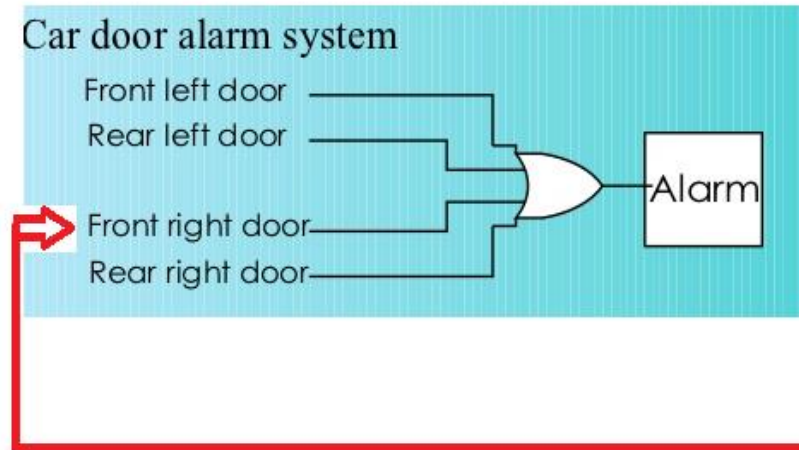


3-Input OR Gate



4-Input OR Gate

Application of an OR gate

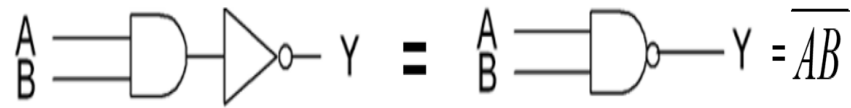


If one or more doors are "ON"/ open,
the car alarm is activated

OR gate in a car door alarm system

Basic Logic Gates (NAND gate)

- NAND is a NOT – AND gate
- Symbol:



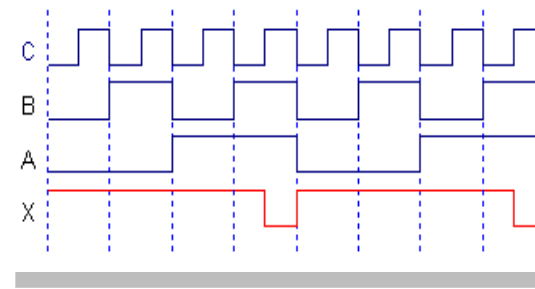
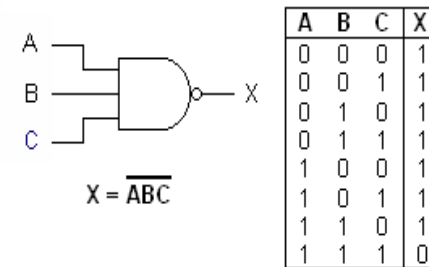
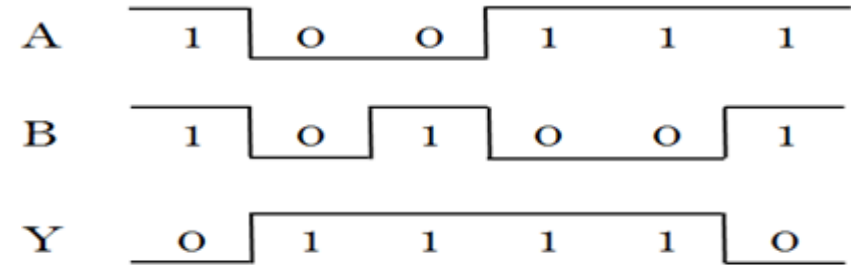
- Truth table:

Input		Output
A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

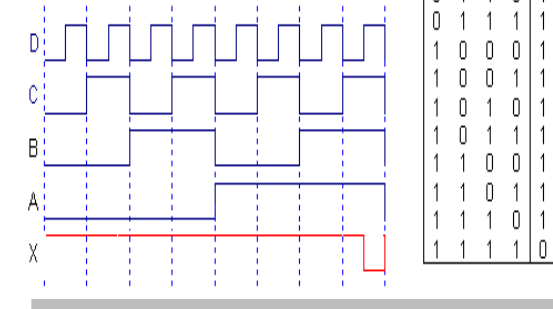
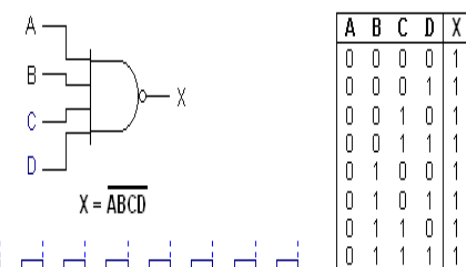
SUMMARY :

- The output is **high (1)** if one or more inputs are **low (0)**
- The output is **low (0)** if all inputs are high (1)

- Timing diagram



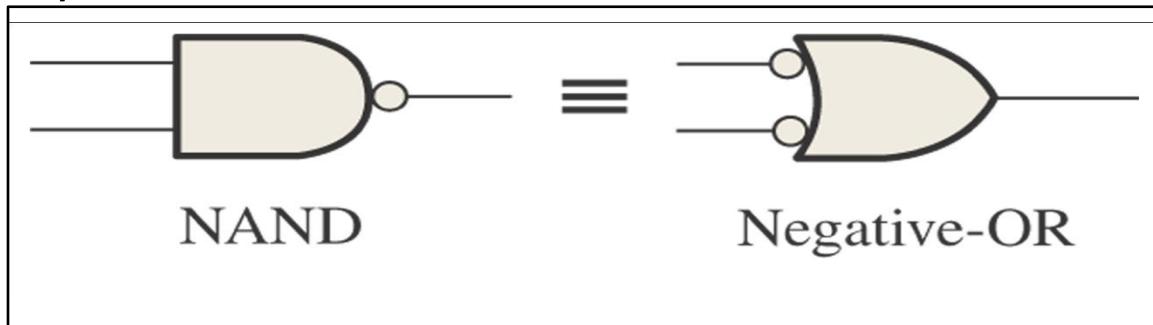
3-Input NAND Gate



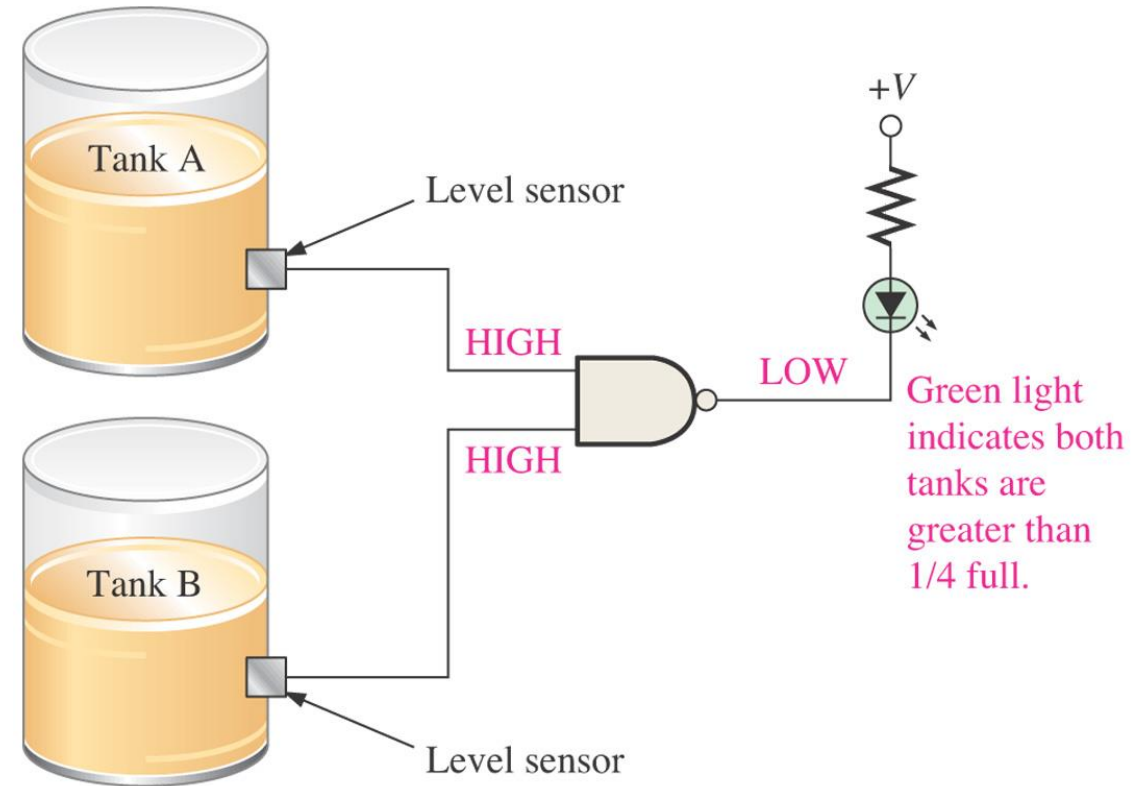
4-Input NAND Gate

Basic Logic Gates (NAND gate)

- Negative-OR Equivalent to a NAND Gate
 - For a 2-input NAND gate performing a negative-OR operation, output X is HIGH if either input A or input B is LOW, or if both A and B are LOW.
- Symbol



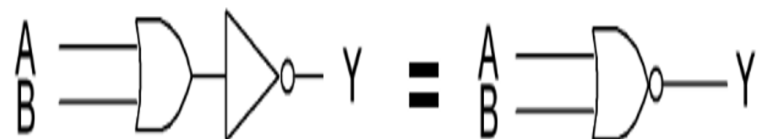
- Application of a NAND gate



Level indicator using NAND gates

Basic Logic Gates (NOR gate)

- NOR is a NOT – OR gate
- Symbol:



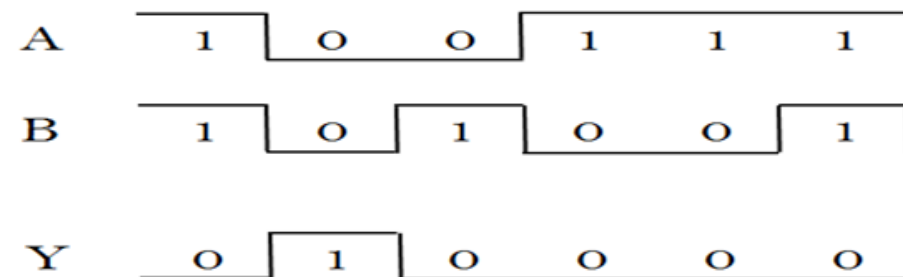
- Truth table:

Input		Output
A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

SUMMARY :

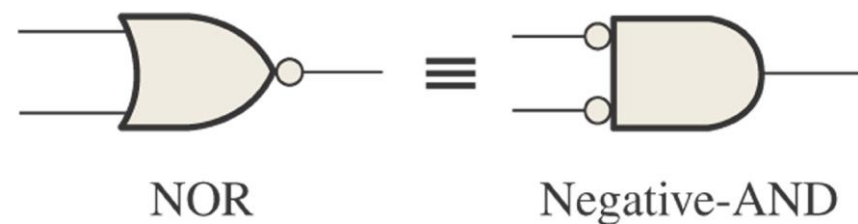
- The output is **high (1)** if all inputs are **low (0)**
- The output is **low (0)** if one or more inputs are high (1)

- Timing diagram



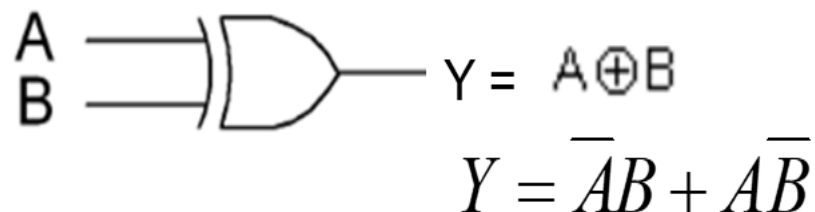
- Negative-AND Equivalent to a NOR Gate

- For a 2-input NOR gate performing a negative-AND operation, output X is HIGH if both input A and input B are LOW.

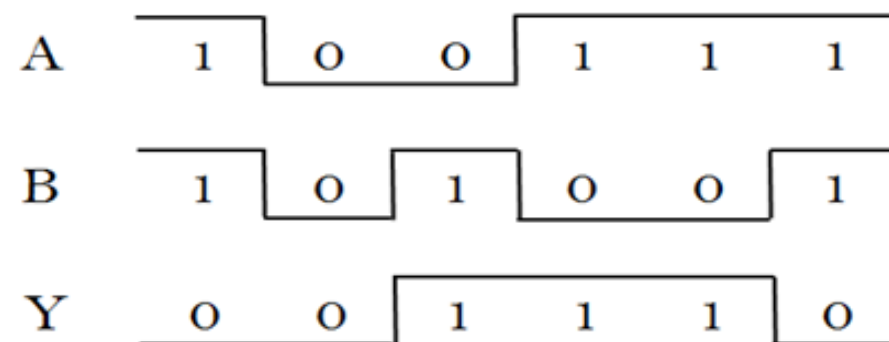


Basic Logic Gates (Exclusive-OR, XOR gate)

- Symbol:



- Timing diagram



- Truth table:

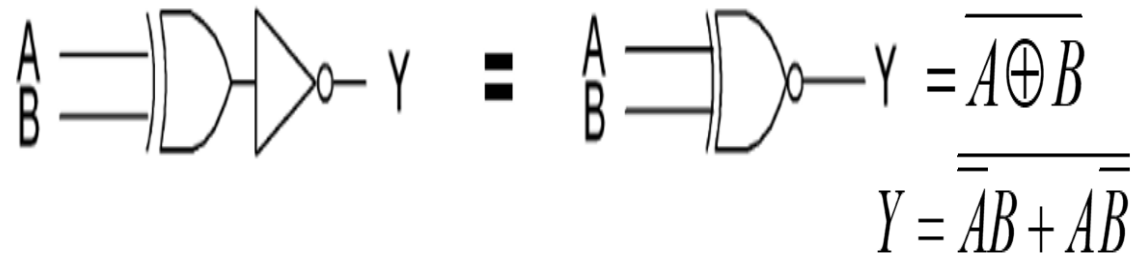
Input		Output
A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

SUMMARY :

- The output is **high (1)** if both inputs are not similar
- The output is **low (0)** if both inputs are similar

Basic Logic Gates (Exclusive-NOR, XNOR gate)

- Symbol:



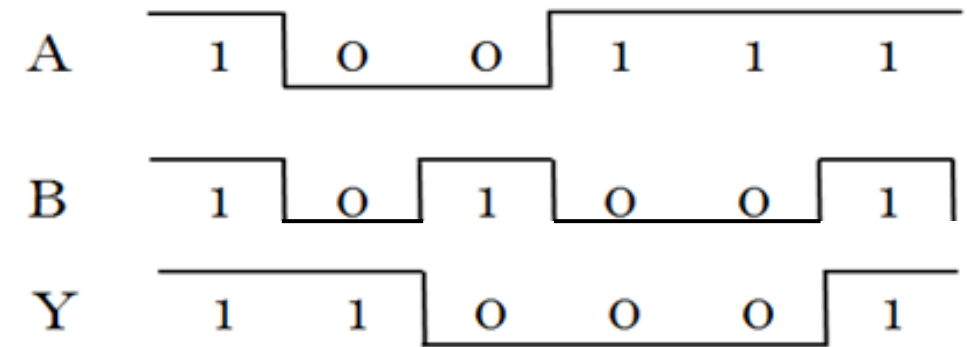
- Truth table:

Input		Output
A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

SUMMARY :

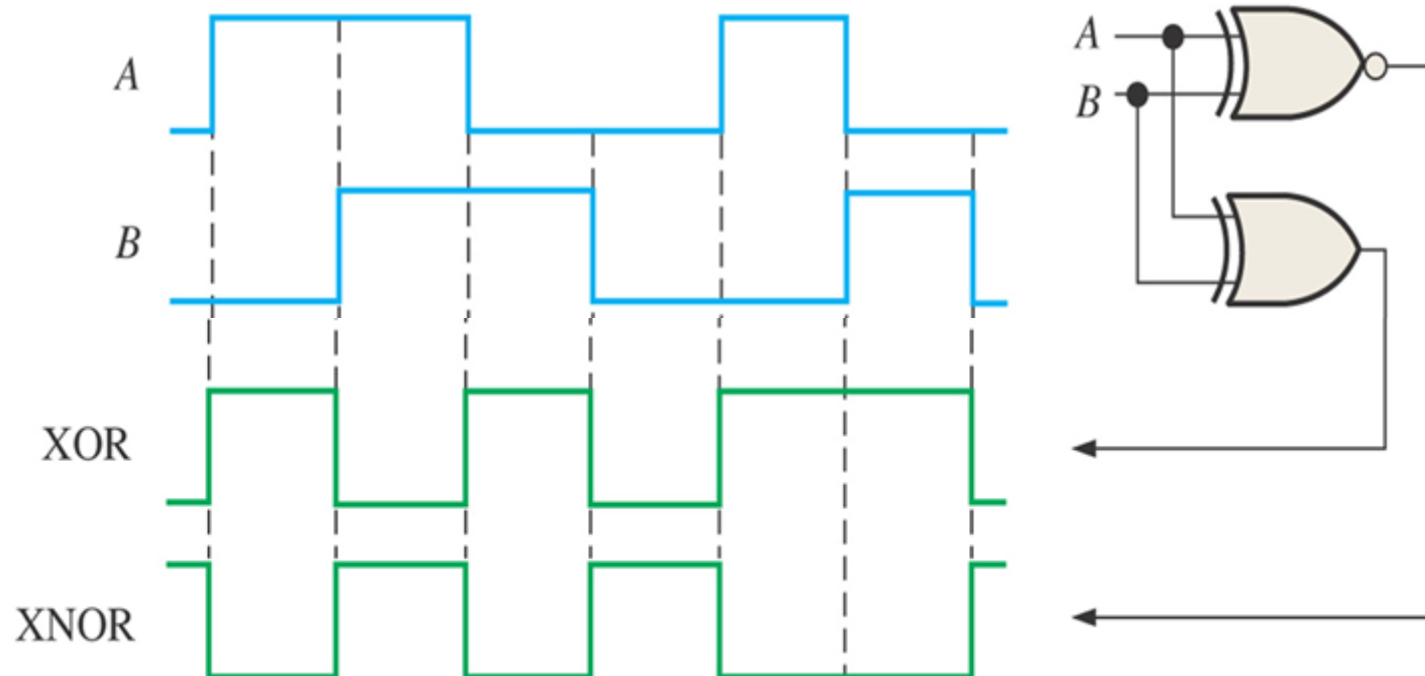
- (i) The output is **high (1)** if both inputs are similar
- (ii) The output is **low (0)** if both inputs are not similar

- Timing diagram



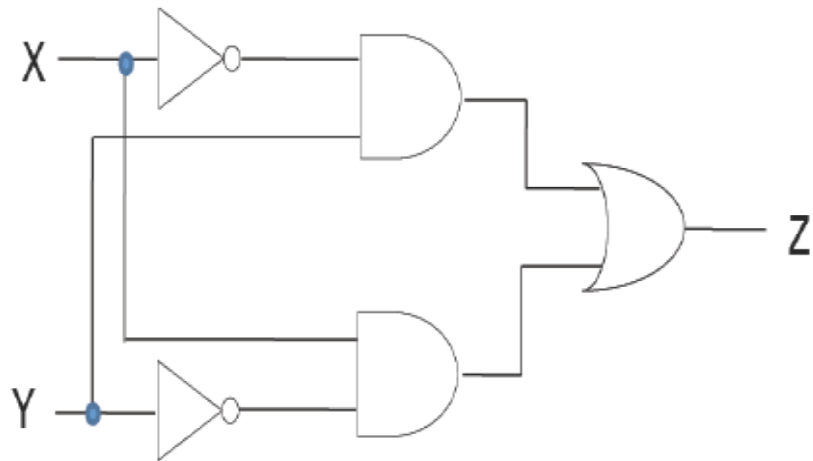
XOR and XNOR

Example 1: Sketch the output of both the gates

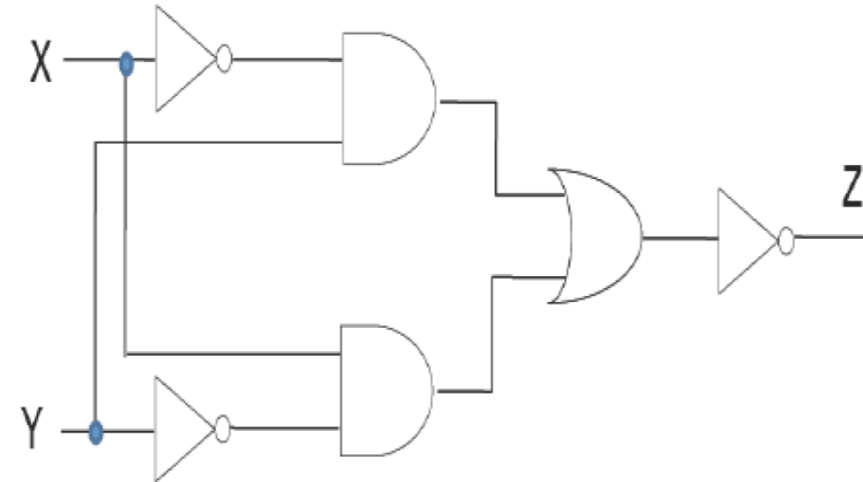


XOR and XNOR

Example 2: Sketch the logic gate XOR and XNOR using basic AOI gate(s)



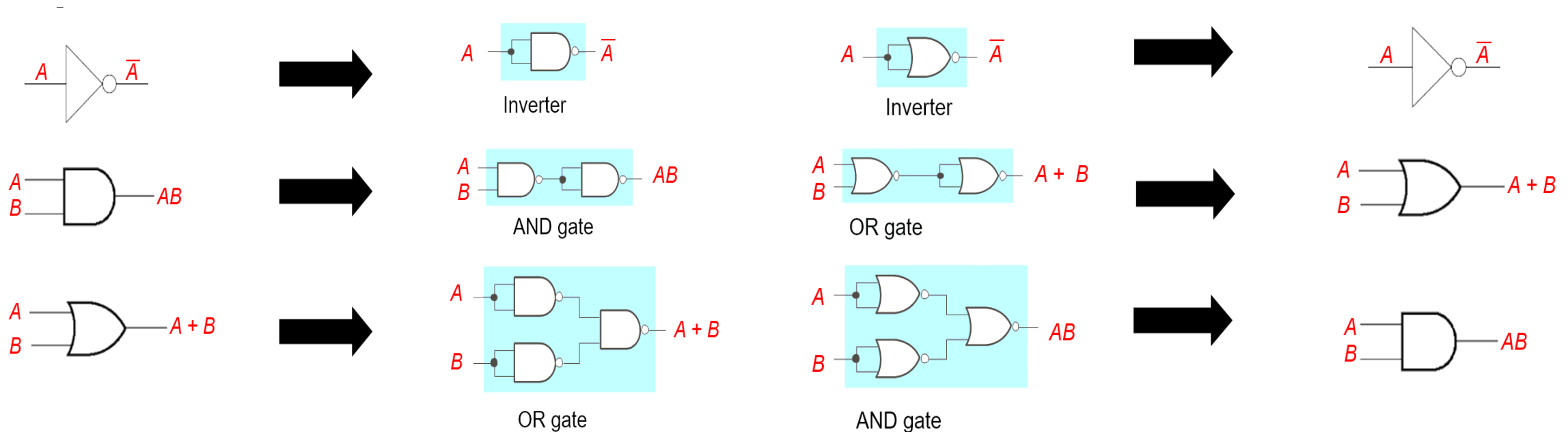
XOR



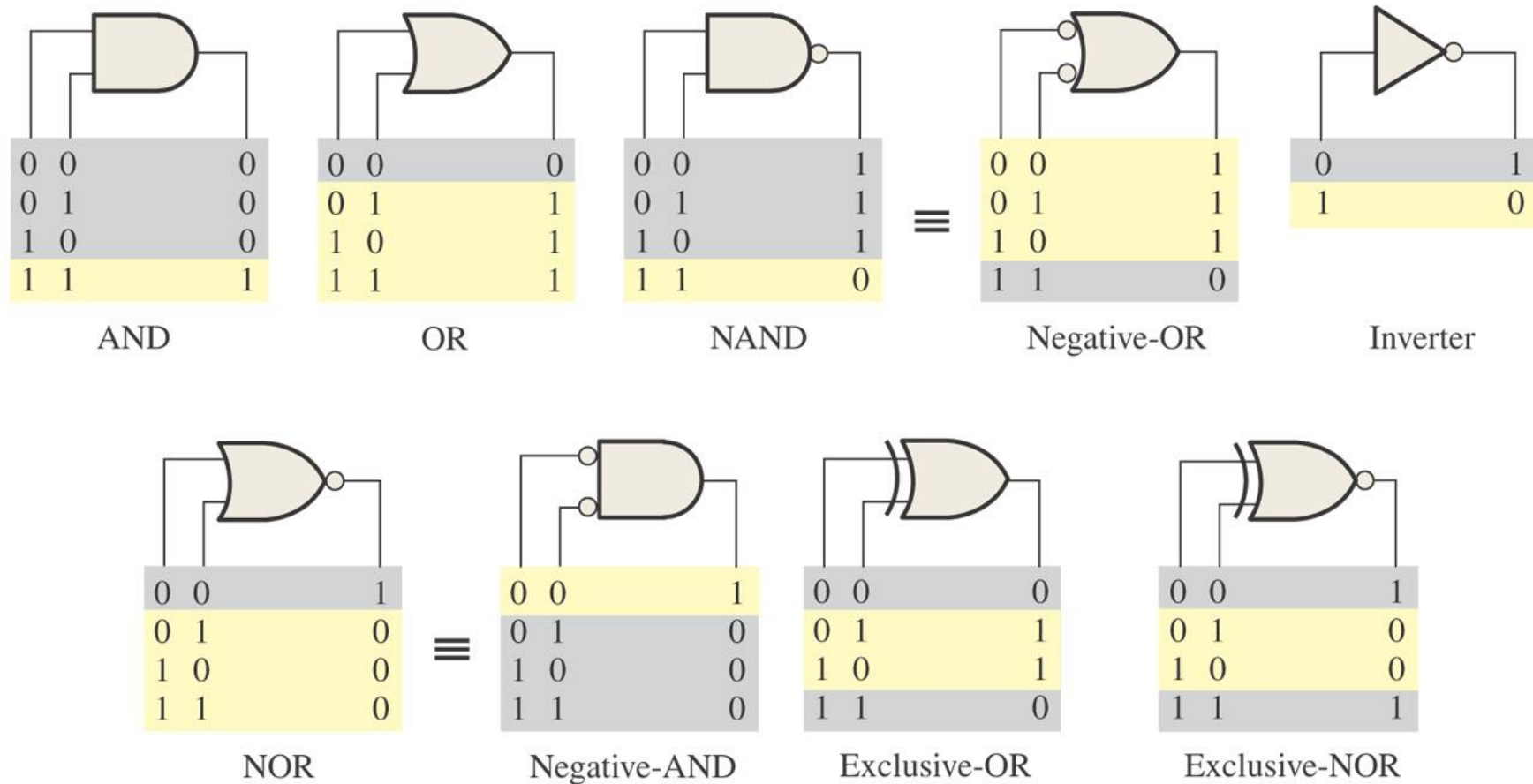
XNOR

NAND & NOR gates

- The NAND and NOR gates are also known as **universal** gates because they can be used to produce the other basic Boolean functions.



SUMMARY : Symbols and truth tables for various logic gates (limited to 2 inputs)

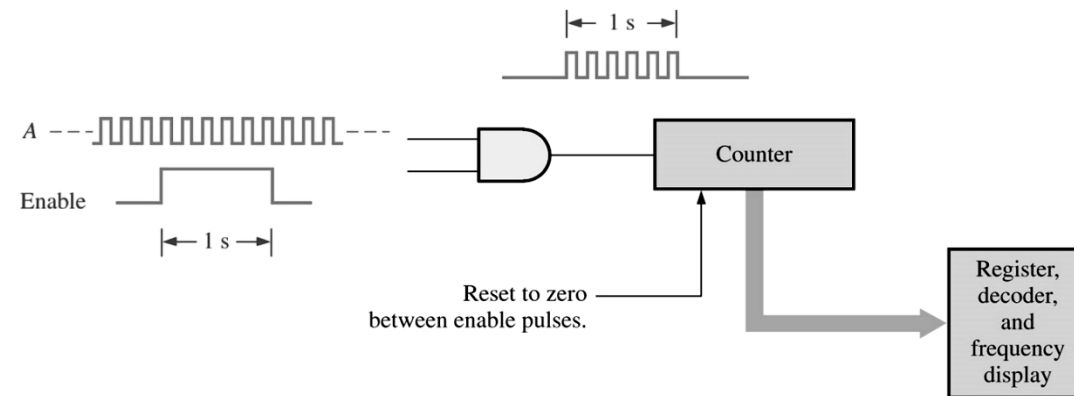


Note: Active states are shown in yellow.

Application of a basic logic gate(s)

Example 3: As an enable and disable function

A circuit is used to measure the frequency of waveform A. (*Frequency $\Rightarrow$ rate of occurrence i.e. how many occurrences per second*). The AND gate is enabled for 1 s for waveform A pulses to pass through and counted by the counter. Then the AND gate is disabled by de-asserting enable signal. No pulses to the counter. Signal A is blocked / disabled by the AND gate. The counter counts the number of pulses per second and produces a binary output that goes to a decoding and display circuit to produce readout of the frequency.

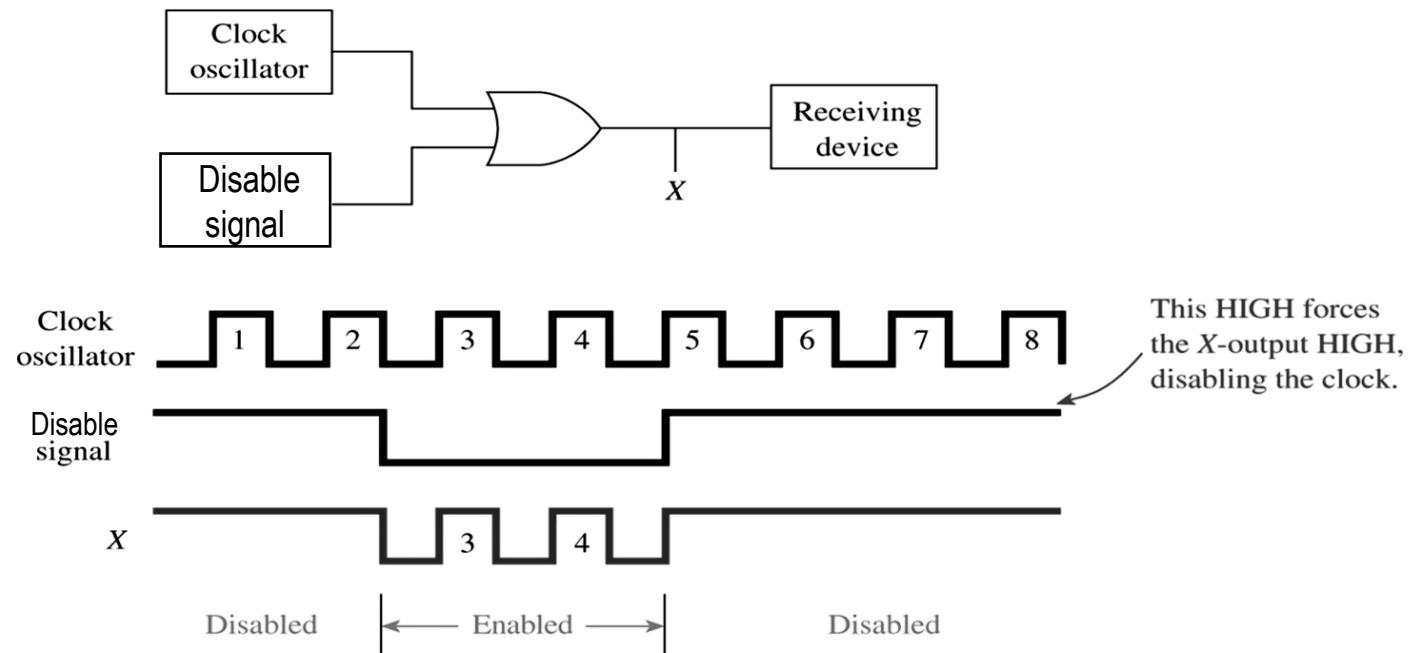


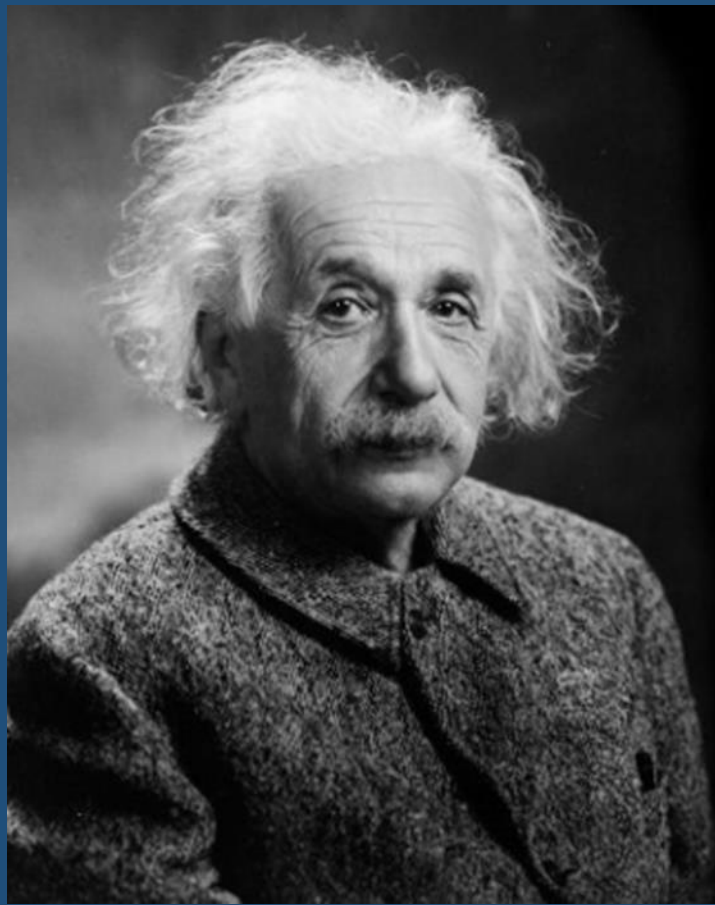
The **enable pulse** is repeated at certain intervals for new count. If the frequency of signal A changes, a new value will be displayed. Between enable pulses, the counter is reset to zero to ready for a new count.

Application of a basic logic gate(s)

Example 4: As an enable and disable function

A 2-input OR gate is used to enable or disable a clock signal. Instead of using an enable signal name, it is more appropriate to use the disable signal name. When the disable signal is asserted, the OR output stays HIGH – no clock signal appears on the OR output. When the disable signal is de-asserted, the clock signal appears on the OR output.





“Creativity is
intelligence
having fun”

– Albert Einstein



Q & A

Thank you